

Strategic coexistence theory for evolutionary games

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Abstract

Evolutionary game theory and ecological coexistence theory both seek to predict the outcome of competition between biological entities, be they strategies or species, but the two fields have relied on largely separate approaches. Replicator equations provide a foundation for predicting the outcome of strategic competition, yet they do not explicitly decompose the mechanisms underlying these outcomes. Inspired by modern coexistence theory from community ecology, we develop Strategic Coexistence Theory (SCT), which translates payoff-based invasion conditions into strategic niche differences and fitness differences between any two competing strategies described by classic replicator dynamics. SCT recovers the classic classification of two-strategy games, distinguishing competitive exclusion, coexistence, and priority effects within a shared niche–fitness difference space. Applying SCT to five mechanisms for the evolution of cooperation further reveals that these mechanisms promote cooperation through distinct dynamical routes: kin selection, network reciprocity, and group selection primarily reduce fitness differences, whereas direct and indirect reciprocity destabilize competition and generate priority effects. Finally, applying SCT to microbial public-goods competition shows that nonlinear microbial growth can both stabilize and equalize competition between cooperators and defectors, allowing coexistence. Together, these results show that SCT provides a common language for comparing coexistence mechanisms across evolutionary games.

Significance statement

Evolutionary game theory provides a powerful framework for predicting the outcome of competition between strategies, including the dominance of one strategy, coexistence of multiple strategies, and bistability. However, a common framework for comparing how different mechanisms promote strategy coexistence remains elusive, even though stable polymorphisms, such as the coexistence of cooperators and defectors, are widely observed across biological systems. To address this gap, we developed Strategic Coexistence Theory (SCT), which extends Modern Coexistence Theory from community ecology to evolutionary games. Applying SCT to classic games, mechanisms of cooperation, and microbial public-goods competition shows

that seemingly similar evolutionary outcomes can arise through distinct mechanisms. This approach links ecological theory with replicator dynamics, providing a complementary tool for analyzing evolutionary games.

Introduction

Understanding the mechanisms that promote the evolutionary stability of competing strategies is a fundamental aim in evolutionary biology (Lewontin, 1961; Slobodkin, 1964; Smith and Price, 1973; Taylor and Jonker, 1978). To this end, evolutionary game theory (EGT) and replicator dynamics have provided key foundations, allowing us to predict the outcome of strategic competition through payoff inequalities, invasion criteria, and stability analysis (Schuster and Sigmund, 1983; Hofbauer and Sigmund, 1998; Cressman, 2003; Cressman and Tao, 2014). Recently, there has been growing interest in characterizing similarities between replicator dynamics and the dynamics of competing species (Gokhale and Traulsen, 2025; Tarnita and Traulsen, 2025; Allesina, 2026), building on the classical mathematical equivalence between replicator equations and generalized Lotka–Volterra models (Hofbauer, 1981; Page and Nowak, 2002). In particular, Tarnita and Traulsen (2025) recently argued that a closer synthesis between ecology and evolutionary game theory can provide new insights into ecological and evolutionary assembly, competition, and coexistence. This renewed perspective between ecological and evolutionary dynamics suggests that evolutionary games may be interpreted not only as payoff-based competition but also as problems of strategic coexistence.

Interestingly, even a simple two-strategy game can yield heterogeneous outcomes that mirror outcomes in community ecology (Hauert, 2002; Szabó and Fath, 2007; Cooney, 2020; Tarnita and Traulsen, 2025): the dominance of a single strategy (prisoner’s dilemma and harmony games), coexistence (hawk–dove games), and bistability (stag-hunt games). Such differences in conditions that promote evolutionary stability and dominance of a single strategy and those that permit strategy coexistence indicate that EGT can be understood from the perspective of coexistence theory (Motro, 1991; Doebeli et al., 2004; Hauert and Doebeli, 2004; Doebeli and Hauert, 2005; Gore et al., 2009; Souza et al., 2009; Archetti and Scheuring, 2011; Steidinger and Bever, 2014; Tarnita and Traulsen, 2025). In fact, stable polymorphisms of evolutionary strategies are observed in microbial and other biological systems (Gore et al., 2009; Cordero et al., 2012; Kocher et al., 2018; Pollak et al., 2021; Lin et al., 2024), but coexistence-theoretic perspective has often remained implicit. Recasting game dynamics as a coexistence problem immediately raises the following question: when two strategies coexist, does the corresponding mechanism stabilize or equalize competition (Chesson, 2000)? While EGT allows us to predict whether two strategies can coexist or not, a common framework for comparing coexistence mechanisms between competing strategies remains elusive.

Modern coexistence theory (MCT) from community ecology presents a particu-

larly useful framework for quantifying coexistence mechanisms (Chesson, 2000; Adler et al., 2007; Godoy et al., 2014; Godoy and Levine, 2014; Kraft et al., 2015; Adler et al., 2026). MCT predicts that coexistence of two competing entities, be they species or strategies, requires stabilizing niche differences to overcome average fitness differences (Chesson, 2000; Adler et al., 2007; Kraft et al., 2015). In community ecology, stabilizing niche differences represent differences in factors that limit the growth of competing species, causing each species to limit itself more than it limits its competitor (Chesson, 2000). In EGT, a strategy’s niche can be interpreted as strategic environments defined by the composition of competing strategies and the resulting interactions (Taylor and Jonker, 1978; Cressman and Tao, 2014; Tarnita and Traulsen, 2025). Thus, stabilizing niche differences arise when each strategy is favored by a different strategic environment, such that the spread of each strategy creates conditions that limit its own relative growth and favor the recovery of its competitor. In contrast, fitness differences represent differences in the ability of each entity to compete under the limiting environments created by its competitor (Chesson, 2000). This differs from payoff differences in EGT, which depend on the underlying strategy frequency and therefore combine both stabilizing effects caused by changes in the strategic environment and residual fitness differences that favor one strategy over another (Taylor and Jonker, 1978; Cressman and Tao, 2014; Tarnita and Traulsen, 2025).

So far, MCT has been primarily utilized for understanding species coexistence, but recent work showed that MCT can be extended beyond species competition, especially for studying pathogen strain competition, allowing one to directly quantify niche and fitness differences of competing entities, be they species or strains (Sieben et al., 2022; Park et al., 2024, 2026). Inspired by this effort, we present Strategic Coexistence Theory (SCT) for predicting coexistence and exclusion of competing strategies in EGT by decomposing invasion growth rates into strategic niche and fitness differences. We begin by introducing SCT, which can be applied to any generic system of EGT described by replicator equations. We show that SCT can recover outcomes from classic, two-strategy games, providing a reinterpretation of classic results from coexistence-theoretic perspective. We then apply SCT to comparing five mechanisms for the evolution of cooperation, revealing that direct and indirect reciprocity destabilize competition between cooperators and defectors, thus pushing the system towards priority effect regime. Finally, we consider a case study of microbial cooperation, where nonlinear microbial growth can both stabilize and equalize competition between cooperation and defection, allowing coexistence of strategies. Together, these examples show that SCT provides a common, ecological language for comparing the mechanisms underlying strategic evolution.

Strategic coexistence theory

Here, we present a brief derivation of strategic coexistence theory (SCT) (Figure 1). To do so, we begin with a classic replicator equation, which allows us to model changes in the frequency of competing strategies based on their payoffs (Schuster and Sigmund, 1983; Hofbauer and Sigmund, 1998; Cressman, 2003; Cressman and Tao, 2014):

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (1)$$

where x_i represents the frequency of strategy i , $\pi_i(\mathbf{x})$ represents the payoff of strategy i for a given strategy distribution $\mathbf{x}$, and $\bar{\pi}(\mathbf{x}) = \sum_i \pi_i(\mathbf{x})x_i$ represents the average payoff. Since strategy frequencies must sum to 1 (i.e., $\sum_i x_i = 1$), a single-strategy equilibrium dominated by strategy i can be described by a unit vector $\mathbf{e}_i$, where all elements are equal to zero except for the i -th element. We use these single-strategy equilibria to define niche and fitness differences between two competing strategies i and j .

Because payoffs in many games can be negative, payoff ratios can be difficult to interpret. Thus, we work on a positive multiplicative scale by exponentiating the payoff terms:

$$W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x})). \quad (2)$$

This exponential transformation is convenient because it preserves the payoff ordering and does not alter the invasion conditions. Therefore, this allows us to compare strategic environments through ratios, as in modern coexistence theory. For convenience, we refer to $W_i(\mathbf{x})$ as the multiplicative payoff of strategy i .

First, we define the niche overlap ρ between strategies i and j as the ratio between the geometric mean of each strategy's multiplicative payoff in the strategic environment that it creates when common, $\sqrt{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}$, and the geometric mean of each strategy's multiplicative payoff in the environment generated by its competitor, $\sqrt{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}$. The corresponding niche difference is then given by

$$1 - \rho = 1 - \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (3)$$

When $1 - \rho > 0$, each strategy performs better, on average, in the environment generated by its competitor than in the environment generated by itself. In other words, intraspecific competition is stronger than interspecific competition. When $\rho > 1$, we interpret the resulting negative niche difference, $1 - \rho < 0$, as a destabilizing effect corresponding to positive frequency dependence, meaning that the strategy that is already common is favored.

Second, we define the fitness difference as the ratio between the geometric means of the multiplicative payoff of each strategy across the two single-strategy environments:

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (4)$$

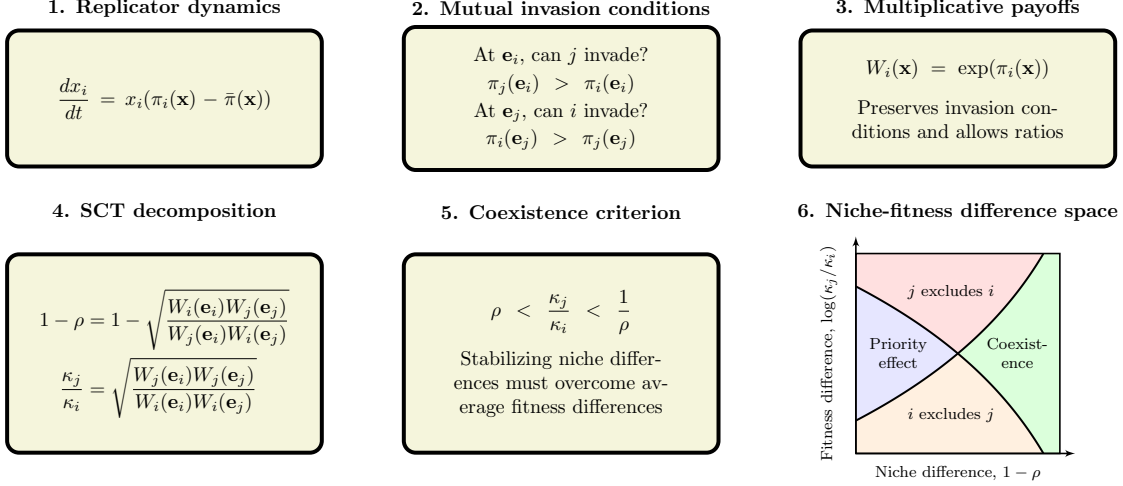


Figure 1: **A schematic diagram summarizing strategic coexistence theory.** Strategic coexistence theory decomposes pairwise replicator dynamics into stabilizing and equalizing mechanisms. Mutual invasion conditions can be rewritten on a multiplicative payoff scale, allowing us to derive strategic niche difference $1 - \rho$ and fitness difference κ_j/κ_i based on multiplicative payoff ratios. Then, the inequality $\rho < \kappa_j/\kappa_i < 1/\rho$ determines mutual invasion. We note that the exact values of niche and fitness differences are sensitive to payoff scaling, while dynamical outcomes are still preserved.

150 In other words, strategy j has an average competitive advantage over strategy i
 151 ($\kappa_j/\kappa_i > 1$) when it performs better, on average, across the environments generated
 152 by itself and its competitor.

153 Together, for any two strategies described by the classic replicator equation, mu-
 154 tual invasion requires stabilizing niche differences to overcome average fitness differ-
 155 ences. Specifically, the condition for mutual invasion can be rewritten as (Chesson,
 156 2000):

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (5)$$

157 This inequality applies to any pairwise comparison between two strategies described
 158 by the classic replicator equation because ρ and κ_j/κ_i can be calculated directly from
 159 ratios of multiplicative payoffs. This decomposition not only allows us to understand
 160 how different mechanisms contribute to the coexistence or exclusion of competing
 161 strategies but also provides shared axes for comparing outcomes of different games.
 162 Detailed derivations are provided in Supplementary Text S1.

163 We note that the absolute magnitudes of niche and fitness differences are sen-
 164 sitive to payoff scaling because they are defined on an exponentiated payoff scale.
 165 Therefore, while SCT allows us to compare different games on shared axes, we can-
 166 not compare the exact values of niche and fitness differences across different games.

167 Nonetheless, as we show in Supplementary Text S2, the qualitative outcome of strate-
 168 gic competition under SCT is preserved because payoff rescaling preserves payoff
 169 orderings. Therefore, we focus on comparing qualitative outcomes from SCT, in-
 170 cluding directions of change in niche and fitness difference and resulting dynamical
 171 regimes, and refrain from making any comparisons of magnitude of niche and fitness
 172 difference values.

173 Classic two-strategy games

174 As an illustrative example, we begin with classic two-strategy games (Hauert, 2002;
 175 Szabó and Fath, 2007; Cooney, 2020; Tarnita and Traulsen, 2025), which can be
 176 described by the following payoff matrix (Figure 2A):

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (6)$$

177 Here, the first and second rows represent strategies 1 and 2, respectively, and the
 178 columns represent the strategy of the interacting opponent. Writing x_1 to denote
 179 the frequency of strategy 1, the expected payoffs are

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (7)$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (8)$$

180 Thus, the dynamics can be written as

$$\frac{dx_1}{dt} = x_1(1 - x_1) [(R - T)x_1 + (S - P)(1 - x_1)]. \quad (9)$$

181 The signs of $T - R$ and $S - P$ determine whether each strategy can invade the
 182 other when rare. Specifically, strategy 2 can invade strategy 1 when $T > R$, whereas
 183 strategy 1 can invade strategy 2 when $S > P$. These two invasion conditions partition
 184 the game space into four regimes (Figure 2B): prisoner’s dilemma ($T > R > P >$
 185 S), hawk–dove game ($T > R, S > P$), stag–hunt game ($R > T, P > S$), and
 186 harmony game ($R > T, S > P$). In standard EGT, these outcomes are obtained by
 187 analyzing the stability of equilibrium conditions. For simple games, this analysis is
 188 straightforward.

189 A coexistence-theoretic approach recovers the same classification but provides
 190 a different interpretation (Figure 2C). For a two-strategy game determined by this
 191 payoff matrix, the niche difference and fitness difference are given by the following
 192 set of equations (Supplementary Text S3):

$$1 - \rho = 1 - \exp\left(\frac{R + P - S - T}{2}\right), \quad (10)$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (11)$$

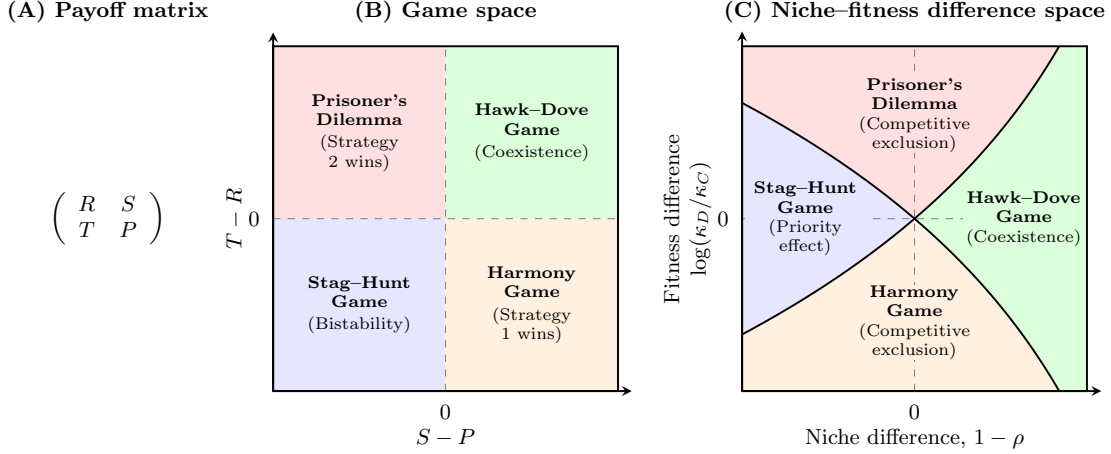


Figure 2: **Coexistence-theoretic interpretation of classic two-strategy games.** (A) General payoff matrix for a two-strategy game, where rows represent the focal strategy and columns represent the strategy of the interacting opponent. (B) Classification of four classic games based on the payoff differences $T - R$ and $S - P$. Prisoner's dilemma and harmony games lead to the dominance of one strategy; hawk-dove games allow coexistence; and stag-hunt games result in bistability. (C) Mapping of the same four games onto niche-fitness difference space. Hawk-dove games correspond to the coexistence regime, where stabilizing niche differences overcome fitness differences. Prisoner's dilemma and harmony games correspond to competitive-exclusion regimes, whereas stag-hunt games correspond to the priority-effect regime. Negative values of $1 - \rho$ indicate positive frequency dependence and destabilization. $R > P$ is assumed throughout, but this does not affect coexistence outcome in any way.

As shown above, mutual invasion requires $\rho < \kappa_2/\kappa_1 < 1/\rho$, which is equivalent to $T > R$ and $S > P$. This corresponds to the hawk-dove game, in which each strategy can invade its competitor when rare and the two strategies coexist. The remaining games also have natural interpretations in niche-fitness difference space. Prisoner's dilemma and harmony games represent competitive exclusion regimes, where one strategy has a fitness advantage that exceeds niche differences. In contrast, stag-hunt games represent a priority-effect regime, where neither strategy can invade the other and the outcome depends on initial conditions. Therefore, coexistence theory not only allows us to recover the classic results from two-strategy games but also provides new interpretations for these results in terms of niche and fitness differences. Building on this, we next apply this framework to comparing core mechanisms of cooperation.

Comparisons of mechanisms for the evolution of cooperation

Understanding mechanisms that promote the evolution of cooperation is a central goal in evolutionary biology (West et al., 2007; Fletcher and Doebeli, 2008; Clutton-Brock, 2009; Kingma et al., 2014; Rakoff-Nahoum et al., 2016; Hilbe et al., 2018; Apicella and Silk, 2019; West et al., 2021; Lee et al., 2022; Palmer and Foster, 2022; Michel-Mata et al., 2024). Here, we compare five classic mechanisms for the evolution of cooperation using SCT (Nowak, 2006) and ask how they differ from the perspective of SCT.

Specifically, we consider the following mechanisms: kin selection, direct reciprocity, indirect reciprocity, network reciprocity, and group selection (Figure 3A). Kin selection favors cooperation when two interacting individuals are genetic relatives, where the genetic relatedness r must be greater than the cost-to-benefit ratio: $r > c/b$ (Hamilton, 1964). Direct reciprocity favors cooperation when current cooperation may promote future cooperation by the same individual through repeated encounters (Trivers, 1971; Axelrod and Hamilton, 1981). In this case, the probability of repeated interaction (i.e., “next round”) must be greater than the cost-to-benefit ratio: $w > c/b$. Indirect reciprocity favors cooperation when current cooperation may promote future cooperation by different individuals through reputation (Nowak and Sigmund, 1998; Wedekind and Milinski, 2000; Nowak and Sigmund, 2005). Thus, the social acquaintanceship (i.e., the probability of knowing someone’s reputation) must be greater than the cost-to-benefit ratio: $q > c/b$. Network reciprocity favors cooperation via clustering of cooperators, requiring the benefit-to-cost ratio to be greater than the average number of neighbors: $b/c > k$ (Nowak and May, 1992; Lieberman et al., 2005). Finally, group selection favors cooperation via multilevel selection, where groups with more cooperators exhibit faster growth at the group level even if defectors reproduce faster at the individual level (Traulsen and Nowak, 2006). This requires $b/c > 1 + (n/m)$, where n is the maximum group size and m is the number of groups. All of these mechanisms can be described by a simple 2×2 payoff matrix (Nowak, 2006), from which the corresponding niche and fitness differences can be computed using Eq. (10) and Eq. (11). Details on niche and fitness difference calculations are presented in Supplementary Text S4.

First, the prisoner’s dilemma provides a baseline scenario, where defection is evolutionarily stable (Figure 3A). From the perspective of SCT, there is complete niche overlap, and therefore defection competitively excludes cooperation (Figure 3B). Building on this, SCT shows that five mechanisms promote evolutionary stability of cooperation via different dynamical routes (Figure 3B). For example, kin selection allows the fitness of cooperation to increase as genetic relatedness r increases (Figure 3B). However, it does not affect the niche difference between cooperation and defection. Similarly, network reciprocity and group selection modulate the fitness of cooperation without causing any changes in strategic niches. In contrast, direct and indirect reciprocity destabilize competition while increasing the fitness of coop-

(A)

Prisoner's dilemma	Kin selection
$\begin{pmatrix} b-c & -c \\ b & 0 \end{pmatrix}$	$\begin{pmatrix} (b-c)(1+r) & br-c \\ b-r & 0 \end{pmatrix}$
Direct reciprocity	Indirect reciprocity
$\begin{pmatrix} (b-c)/(1-w) & -c \\ b & 0 \end{pmatrix}$	$\begin{pmatrix} b-c & -c(1-q) \\ b(1-q) & 0 \end{pmatrix}$
Network reciprocity	Group selection
$\begin{pmatrix} b-c & H-c \\ b-H & 0 \end{pmatrix}$	$\begin{pmatrix} (b-c)(m+n) & (b-c)m-cn \\ bn & 0 \end{pmatrix}$

(B)

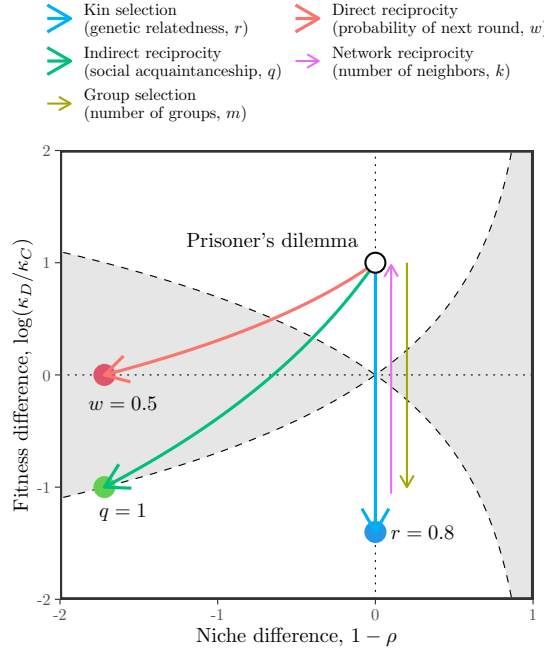


Figure 3: **Qualitative comparisons of five mechanisms for the evolution of cooperation using SCT.** (A) Payoff matrices for the baseline scenario (prisoner's dilemma) and five mechanisms for the evolution of cooperation (Nowak, 2006). Prisoner's dilemma is characterized by the benefit for the recipient b and the cost for the donor c . Network reciprocity is parameterized based on $H = [(b-c)k - 2c]/[(k+1)(k-2)]$. Parameters for all other mechanisms are described in panel B. (B) Changes in niche and fitness differences driven by each mechanism. Each arrow represents the direction of change as we increase the corresponding parameter, indicated in figure legends. Note that kin selection, network reciprocity, and group selection all result in changes in fitness difference without causing any changes in niche difference. Therefore, we indicate the direction of change for network reciprocity and group selection separately to avoid overlapping arrows. For simplicity, we focus on the effect of number of groups m for group selection; increasing the number of maximum group size n has the opposite effect (Supplementary Text S4). Across all models, we assume $b = 3$ and $c = 1$. Because SCT decompositions depend on positive payoff scaling, exact values of niche and fitness difference (e.g., arrow lengths, slopes, and distances from regime boundaries) should not be compared across mechanisms.

246 eration, thereby pushing the competition outcome towards a priority effect regime.
 247 Equivalently, these two mechanisms promote bistability of defection and cooperation.
 248 SCT further reveals another difference between direct and indirect reciprocity.
 249 Direct reciprocity remains within the priority-effect regime as w reaches 1, whereas

indirect reciprocity reaches the boundary between the priority-effect and competitive-exclusion regimes at $q = 1$ (Supplementary Text S4).

We note that none of these mechanisms allow coexistence of cooperation and defection. In many biological systems, such coexistence is often observed (Gore et al., 2009; Cordero et al., 2012; Kocher et al., 2018; Pollak et al., 2021; Lin et al., 2024). Therefore, we move on to the final example, applying SCT to microbial cooperation, where nonlinear benefits and environmental context can promote coexistence of cooperators and defectors.

Can environmental stochasticity prevent coexistence?

So far, we have focused on simple deterministic games. However, many ecological and evolutionary processes are intrinsically stochastic. Recently, Wang et al. (2025) showed that environmental fluctuations can promote coexistence in games that would otherwise lead to competitive exclusion in a deterministic environment. From the perspective of SCT, this suggests that stochasticity can stabilize competition by promoting the recovery of strategies when rare. Interestingly, Wang et al. (2025) also showed that stochasticity can generate bistable coexistence and multiple internal equilibrium. Thus, although stochasticity can stabilize competition near the boundaries, it can also generate destabilizing frequency dependence in the interior. This immediately raises a question of stochasticity can generate sufficiently strong positive frequency dependence to destabilize competition, thus pushing the system out of the coexistence regime. Here, we use SCT to separate these effects and clarify whether stochasticity stabilizes or destabilizes strategic competition.

Following Wang et al. (2025), we consider a 2×2 game in the presence of environmental noise. The stochastic payoff matrix is given by

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix} + \frac{\zeta}{\sqrt{s}} \begin{pmatrix} \sigma_R & \sigma_S \\ \sigma_T & \sigma_P \end{pmatrix}, \quad (12)$$

where the first matrix represents deterministic payoffs, ζ represents a noise term with zero mean and unit variance, and the second matrix determines the intensity of environmental stochasticity. Assuming weak selection $s \ll 1$, the evolution of two competing strategies in a large population can be described by the following equation:

$$\frac{dx}{dt} = sx(1-x) \left(\pi_C - \pi_D + \left(\frac{1}{2} - x \right) (\sigma_C - \sigma_D)^2 \right), \quad (13)$$

where $\pi_C = Rx + S(1-x)$ and $\pi_D = Tx + P(1-x)$ and $\sigma_C = \sigma_R x + \sigma_S(1-x)$ and $\sigma_D = \sigma_T x + \sigma_P(1-x)$. After rescaling time as $\tau = st$, the invasion growth rates of each strategy when rare can be written as:

$$r_C = S - P + \frac{1}{2}(\sigma_S - \sigma_P)^2, \quad (14)$$

$$r_D = T - R + \frac{1}{2}(\sigma_T - \sigma_R)^2. \quad (15)$$

282 This shows that environmental stochasticity increases the invasion growth rates,
 283 meaning that it is intrinsically stabilizing.

284 Then, an extended SCT decomposition shows that niche overlap and fitness dif-
 285 ference can be written as:

$$\rho = \exp \left(\frac{R + P - S - T}{2} - \frac{(\sigma_R - \sigma_T)^2 + (\sigma_S - \sigma_P)^2}{4} \right) \quad (16)$$

$$\frac{\kappa_D}{\kappa_C} = \exp \left(\frac{T + P - R - S}{2} + \frac{(\sigma_R - \sigma_T)^2 - (\sigma_S - \sigma_P)^2}{4} \right) \quad (17)$$

286 Writing $\hat{\rho}$ and $\hat{\kappa}_D/\hat{\kappa}_C$ to denote niche overlap and fitness difference in the absence of
 287 stochasticity and taking logs of both sides for convenience, we have:

$$\Delta \log \rho = \log \rho - \log \hat{\rho} = - \frac{(\sigma_R - \sigma_T)^2 + (\sigma_S - \sigma_P)^2}{4} \quad (18)$$

$$\Delta \log \frac{\kappa_D}{\kappa_C} = \log \frac{\kappa_D}{\kappa_C} - \log \frac{\hat{\kappa}_D}{\hat{\kappa}_C} = \frac{(\sigma_R - \sigma_T)^2 - (\sigma_S - \sigma_P)^2}{4}. \quad (19)$$

288 Therefore, stochasticity reduces niche overlap and therefore increases niche difference.
 289 In contrast, stochasticity can either amplify or reduce fitness differences depending on
 290 which strategy's invasion rate is more strongly affected by environmental fluctuations.
 291 Notably, changes in fitness difference driven by environmental stochasticity will be
 292 always bounded by those in niche overlap:

$$\Delta \log \rho \leq \Delta \log \frac{\kappa_D}{\kappa_C} \leq -\Delta \log \rho \quad (20)$$

293 This result implies that environmental stochasticity cannot generate sufficient fitness
 294 difference to push the system out of the coexistence regime.

295 As an example, consider a scenario where the amount of stochasticity is assumed
 296 to be proportional to deterministic payoffs (Wang et al., 2025):

$$\begin{pmatrix} \sigma_R & \sigma_S \\ \sigma_T & \sigma_P \end{pmatrix} = k \begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (21)$$

297 Following previous case studies, we consider four deterministic payoff structures (Fig-
 298 ure 4A): Prisoner's Dilemma (PD), Hawk-Dove Game (HD), Stag-Hunt Game (SH)
 299 and Harmony Game (HG). Across all these examples, stochasticity eventually pushes
 300 the system inside the mutual invasion regime (Figure 4B). Notably, changes in dy-
 301 namical regimes predicted by SCT accurately capture bifurcations occurring at the
 302 boundaries (Figure 4C): for PD, HD, and SH games, both boundary equilibria even-
 303 tually become unstable. Notably, for the SH game, the system begins from the
 304 priority effect regime and enters the competitive exclusion regime before eventually
 305 it enters the mutual invasion regime (Figure 4B). Consequently, $x^* = 1$ (pure co-
 306 operation) becomes unstable first, and then $x^* = 0$ (pure defection) also becomes
 307 unstable.

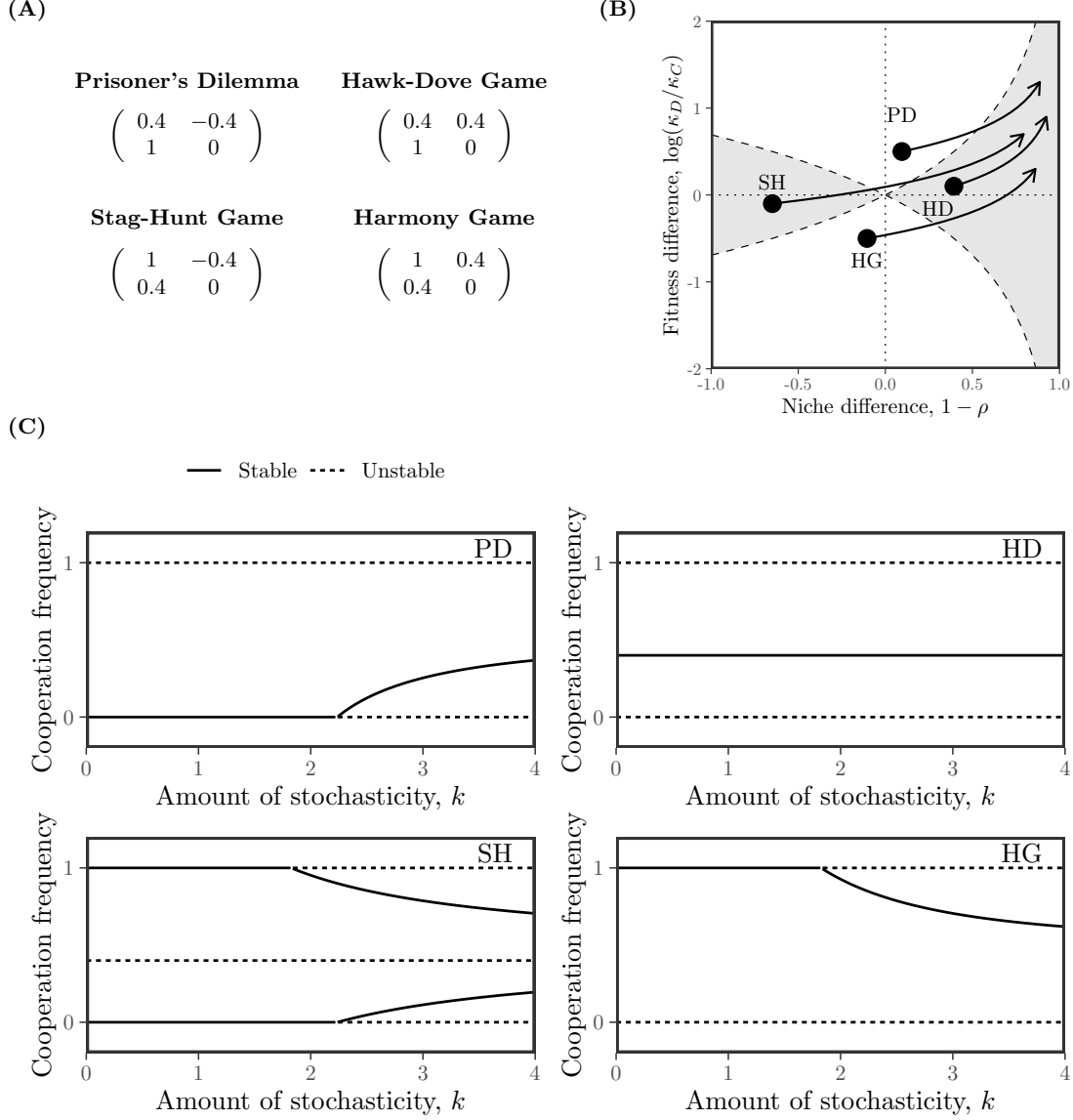


Figure 4: **Effects of environmental stochasticity on strategic competition.** (A) Assumed deterministic payoff structures across four scenarios. Environmental stochasticity is assumed to be proportional to these payoffs, scaled by a term k . (B) Arrows indicate the direction of changes in dynamical regime as the amount of stochasticity k increases. (C) Bifurcation diagrams under each scenario as a function of k . Solid and dashed lines represent stable and unstable equilibrium, respectively.

308 This example also illustrates an important limitation of the original MCT as well
 309 as its extension, SCT. Since MCT and SCT solely rely on boundary conditions, they
 310 cannot predict the dynamics within interior, especially the presence of multiple inter-
 311 nal equilibrium. In our example of the SH game, there is always an unstable internal

equilibrium across all ranges of k we consider (Figure 4C). Thus, when mutual invasion is allowed, the system permits two additional, stable internal equilibria (Figure 4C). In this example, the presence of internal equilibrium creates a contradiction to the traditional interpretation of MCT: when the SH game enters the “competitive exclusion” regime, coexistence becomes possible due to unstable internal equilibrium preventing competitive exclusion. Therefore, while MCT and SCT accurately capture the behavior at the boundary via invasibility, they may not correctly predict whether coexistence is possible.

Microbial cooperation and public goods

Here, we consider a simple, phenomenological model of competition between cooperating and cheating yeast in a cooperative sucrose-metabolism system, where invertase production by cooperators allows cheaters to utilize glucose, a public good released through sucrose hydrolysis (Carlson and Botstein, 1982; Dickinson, 1998; Gore et al., 2009). Specifically, invertase production by cooperators incurs a cost c and generates a total benefit of 1 that is captured privately with efficiency ϵ . Assuming a linear relationship between glucose availability and microbial growth, the payoffs for cooperators π_C and cheaters (defectors) π_D can then be written as

$$\pi_C(f) = \epsilon + f(1 - \epsilon) - c, \quad (22)$$

$$\pi_D(f) = f(1 - \epsilon), \quad (23)$$

where the use of glucose depends on the fraction of cooperators f . The dynamics of cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\pi_C(f) - \pi_D(f)] \quad (24)$$

$$= f(1 - f)(\epsilon - c) \quad (25)$$

For this linear growth model, the payoff difference is fixed to $\epsilon - c$ independent of f (Figure 4A). Therefore, cooperation is favored when efficiency is greater than the cost of cooperation, $\epsilon > c$, whereas defection is favored when the cost is greater than efficiency, $c > \epsilon$ (Gore et al., 2009).

Our framework allows us to reinterpret this result from the perspective of coexistence theory. In particular, the two strategies have no niche difference, corresponding to complete niche overlap, $\rho = 1$. In contrast, the fitness difference is determined by the balance between private efficiency and the cost of cooperation:

$$\frac{\kappa_D}{\kappa_C} = \exp(c - \epsilon). \quad (26)$$

Therefore, this model always predicts competitive exclusion, except in the neutral case $c = \epsilon$. The outcome is determined entirely by the fitness difference between

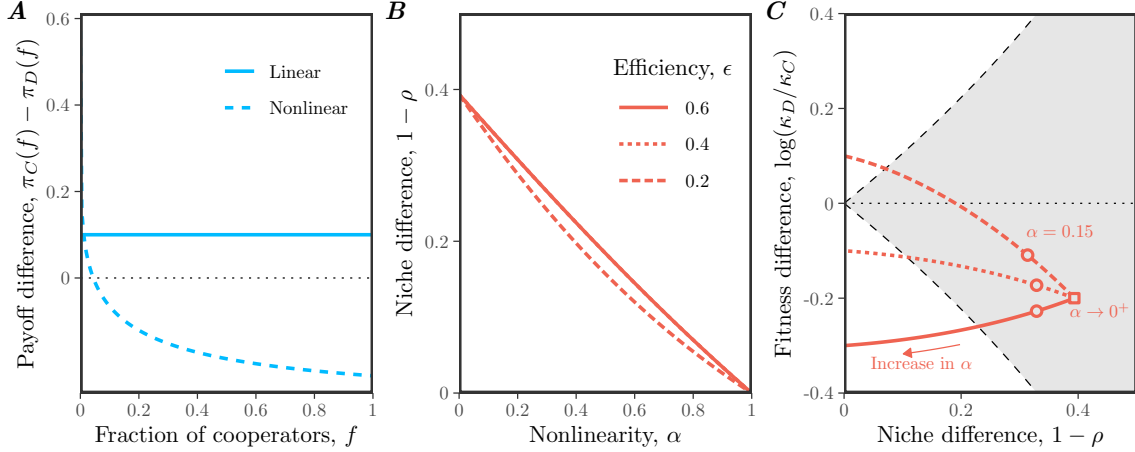


Figure 5: **Effects of nonlinear microbial growth on coexistence of cooperation and defection.** (A) Payoff differences between cooperation and defection for linear and nonlinear models of microbial growth. In panel A, we assumed $\epsilon = 0.4$, $c = 0.3$, and $\alpha = 0.15$ for illustrative purposes. (B) Effect of nonlinearity α on niche difference $1 - \rho$. Each line assumes a different value of efficiency ϵ . Note that lines for $\epsilon = 0.6$ and $\epsilon = 0.4$ are nearly overlapping and are indistinguishable. (C) Effect of nonlinearity α on both niche difference $1 - \rho$ and fitness difference $\log(\kappa_D/\kappa_C)$. The gray shaded area represents the coexistence regime. The square represents the condition when $\alpha \rightarrow 0^+$. The circles represent the condition when $\alpha = 0.15$, which correspond to the experimental relationship between glucose and microbial growth (Gore et al., 2009). In panels B and C, we assumed $c = 0.3$, while allowing ϵ and α to vary.

cooperators and defectors, rather than by stabilizing niche differences. This example resembles the prisoner's dilemma discussed in the previous example (Figure 3).

In real biological systems, the effect of glucose on microbial growth follows a nonlinear relationship:

$$\pi_C(f) = [\epsilon + f(1 - \epsilon)]^\alpha - c, \quad (27)$$

$$\pi_D(f) = [f(1 - \epsilon)]^\alpha, \quad (28)$$

where $\alpha = 0.15$ reflects the experimental relationship between glucose and microbial growth (Gore et al., 2009). Then, the dynamics of cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\{\epsilon + f(1 - \epsilon)\}^\alpha - c - \{f(1 - \epsilon)\}^\alpha]. \quad (29)$$

Under this nonlinear assumption, the payoff difference decreases with the cooperator fraction f , which can promote the coexistence of cooperators and defectors (Figure 4A): cooperators have a competitive advantage when rare, whereas defectors have a competitive advantage when cooperators are abundant.

From a coexistence-theoretic perspective, this means that nonlinear growth can generate sufficient stabilizing niche difference to overcome the fitness difference. In particular, the niche and fitness differences for this model correspond to:

$$1 - \rho = 1 - \exp\left(\frac{1 - \epsilon^\alpha - (1 - \epsilon)^\alpha}{2}\right), \quad (30)$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(c + \frac{(1 - \epsilon)^\alpha - 1 - \epsilon^\alpha}{2}\right). \quad (31)$$

These expressions clarify the separate roles of cost, efficiency, and nonlinearity. For example, the cost of cooperation does not affect niche difference and only affects fitness difference. Instead, niche difference is generated by the concave relationship between glucose availability and microbial growth, which increases from 0 to $1 - \exp(-1/2)$ as α goes from 1 to 0 (Figure 4B,C). In contrast, as $\alpha \rightarrow 0^+$, the fitness difference approaches

$$\log(\kappa_D/\kappa_C) \rightarrow c - \frac{1}{2}, \quad (32)$$

which no longer depends on the private capture efficiency ϵ (Figure 4C). Thus, strong concavity not only generates stabilizing niche differences, but also acts as an equalizing mechanism (Figure 4C). Such stabilizing mechanisms are key ingredients for coexistence of cooperators and defectors.

Discussion

Standard evolutionary game theory and replicator equations provide powerful tools for predicting the outcome of strategy competition, including strategic dominance, coexistence, and bistability (Motro, 1991; Doebeli et al., 2004; Hauert and Doebeli, 2004; Doebeli and Hauert, 2005; Gore et al., 2009; Souza et al., 2009; Archetti and Scheuring, 2011; Steidinger and Bever, 2014; Tarnita and Traulsen, 2025). However, these outcomes are typically understood by analyzing payoff differences within individual games, which makes it difficult to compare coexistence mechanisms across models. Building on Chesson's Modern Coexistence Theory from community ecology (Chesson, 2000), we developed a strategic coexistence theory (SCT) that allows us to decompose strategy competition in terms of stabilizing niche differences and fitness differences. This decomposition provides a unifying framework for comparing different games from the perspective of coexistence theory, allowing us to identify stabilizing and equalizing mechanisms for strategy competition.

To demonstrate the utility of SCT, we considered three case studies. First, the analysis of classic two-strategy games showed that SCT recovers the original classification for these classic games, highlighting the similarities between evolutionary games and community ecology (Hauert, 2002; Szabó and Fath, 2007; Cooney, 2020; Tarnita and Traulsen, 2025). Then, we compared different mechanisms for the evolution of cooperation, which showed that each mechanism promotes cooperation via

different dynamical routes (Nowak, 2006). Kin selection, network reciprocity, and group selection primarily affect the fitness difference between cooperation and defection, whereas direct and indirect reciprocity can destabilize competition and generate priority effects. Finally, our analysis of microbial cooperation showed that coexistence between cooperators and defectors requires more than reducing the fitness cost of cooperation: nonlinear benefits can generate stabilizing niche differences while also reducing fitness asymmetry (Gore et al., 2009). Taken together, these examples show that SCT can provide new insights into ecological underpinnings of strategy competition.

There are several limitations to SCT. First, similar to MCT from community ecology, SCT only allows for comparisons of pairwise strategies and therefore cannot determine the outcomes of games with more than two strategies (Chesson, 2000). Second, the exact values of strategic niche and fitness differences are sensitive to payoff scaling. SCT therefore supports quantitative comparisons only when models share a justified payoff normalization. Without such normalization, comparisons are restricted to qualitative changes in dynamical regimes, directions, and boundary crossings. Third, we focused on classic replicator equations, but Tarnita and Traulsen (2025) recently pointed out that these classic formulations implicitly ignore differences in intrinsic growth rates. In such cases, replicator equations may fail to predict the outcome of competition (Tarnita and Traulsen, 2025), and a full Lotka-Volterra model is needed to describe the abundances, rather than frequencies, of strategies. Thus, SCT inherits the same assumptions as the classic replicator framework: it is most appropriate when strategy competition can be described in terms of relative frequencies alone (Tarnita and Traulsen, 2025). When strategies differ intrinsically in growth, MCT may be required instead to tease apart mechanisms of strategy coexistence. Finally, we have focused on simple examples to illustrate the utility of SCT, but many games involve additional complexities, including population structure (Nowak et al., 2010), stochasticity (Wang et al., 2025; Bodin et al., 2026), and explicit ecological feedback (Tarnita and Traulsen, 2025). Extending SCT to these settings will be necessary to determine how broadly stabilizing and equalizing mechanisms can be compared across evolutionary games.

In conclusion, SCT provides a bridge between evolutionary game theory and community ecology, allowing strategy competition to be understood from the perspective of coexistence theory. Therefore, rather than replacing existing EGT or replicator-dynamics frameworks, SCT provides a complementary tool for teasing apart the coexistence mechanisms underlying evolutionary games. More broadly, our study builds on recent efforts to extend MCT beyond species competition (Park et al., 2024, 2026), providing a foundation for building unifying coexistence theory for competing entities, including species, pathogen strains, and behavioral strategies.

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431 **Competing interests**

432 The authors declare no competing interest.

433 **Data availability**

434 All data and code are stored in a publicly available GitHub repository ([https:](https://github.com/parklab-snu/coexistence-strategy)
435 [//github.com/parklab-snu/coexistence-strategy](https://github.com/parklab-snu/coexistence-strategy)).

436 Supplementary Text

437 S1 Derivation of the strategic coexistence theory

438 To derive strategic coexistence theory (SCT), we begin with replicator equations:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (\text{S1})$$

439 where x_i represents the frequency of strategy i , π_i represents the payoff of strategy
440 i , and $\bar{\pi}$ represents the average payoff. Then, the mutual invasion condition for two
441 competing strategies i and j can be written as

$$\pi_j(\mathbf{e}_j) < \pi_i(\mathbf{e}_j), \quad (\text{S2})$$

$$\pi_i(\mathbf{e}_i) < \pi_j(\mathbf{e}_i). \quad (\text{S3})$$

442 These inequalities will be preserved when we exponentiate the payoffs, $W_i(\mathbf{x}) =$
443 $\exp(\pi_i(\mathbf{x}))$:

$$W_j(\mathbf{e}_j) < W_i(\mathbf{e}_j), \quad (\text{S4})$$

$$W_i(\mathbf{e}_i) < W_j(\mathbf{e}_i). \quad (\text{S5})$$

444 Rearranging, we obtain the following inequality:

$$\frac{W_i(\mathbf{e}_i)}{W_j(\mathbf{e}_i)} < 1 < \frac{W_i(\mathbf{e}_j)}{W_j(\mathbf{e}_j)}. \quad (\text{S6})$$

445 By multiplying $\sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}$ on both sides, we obtain:

$$\sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}} \quad (\text{S7})$$

446 Thus, by defining niche overlap ρ and fitness difference κ_j/κ_i as follows,

$$\rho = \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S8})$$

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S9})$$

447 we obtain the following inequality that defines conditions for mutual invasion and
448 long-term coexistence of two competing strategies:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S10})$$

449 S2 Niche and fitness difference payoff scaling

450 Consider a classic replicator equation:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (\text{S11})$$

451 where x_i represents the frequency of strategy i , π_i represents the payoff of strategy
 452 i , and $\bar{\pi}$ represents the average payoff. In this case, multiplying the payoffs by a
 453 constant $s > 0$ only changes time and does not affect the overall dynamics:

$$\frac{dx_i}{dt} = x_i(\pi'_i(\mathbf{x}) - \bar{\pi}'(\mathbf{x})) \quad (\text{S12})$$

$$= sx_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (\text{S13})$$

454 where $\pi'_i(\mathbf{x}) = s\pi_i(\mathbf{x})$ is the scaled payoff. Then, the multiplicative payoff corresponds
 455 to $W'_i(\mathbf{x}) = \exp(s\pi_i) = W_i(\mathbf{x})^s$.

456 If we try to calculate niche overlap and fitness difference based on the scaled
 457 payoffs, we get:

$$\rho' = \sqrt{\frac{W'_i(\mathbf{e}_i)W'_j(\mathbf{e}_j)}{W'_j(\mathbf{e}_i)W'_i(\mathbf{e}_j)}}, \quad (\text{S14})$$

$$= \left(\sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}} \right)^s, \quad (\text{S15})$$

$$= \rho^s \quad (\text{S16})$$

$$\frac{\kappa'_j}{\kappa'_i} = \sqrt{\frac{W'_j(\mathbf{e}_i)W'_j(\mathbf{e}_j)}{W'_i(\mathbf{e}_i)W'_i(\mathbf{e}_j)}}, \quad (\text{S17})$$

$$= \left(\sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}} \right)^s, \quad (\text{S18})$$

$$= \left(\frac{\kappa_j}{\kappa_i} \right)^s \quad (\text{S19})$$

458 This implies that the exact values of niche and fitness difference are sensitive to
 459 payoff scaling and are not intrinsic properties of a replicator system.

460 Nonetheless, the inequality for mutual invasion condition still holds under payoff
 461 scaling:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho} \iff \rho^s < \left(\frac{\kappa_j}{\kappa_i} \right)^s < \frac{1}{\rho^s}. \quad (\text{S20})$$

462 Furthermore, boundary conditions for complete niche overlap ($\rho = \rho' = 1$) and com-
 463 plete niche difference ($\rho = \rho' = 0$) are preserved. Moreover, the direction of changes
 464 in fitness and niche difference is also preserved under payoff scaling. Specifically, for

parameter θ , we have:

$$\frac{d\rho'}{d\theta} = s\rho^{s-1} \frac{d\rho}{d\theta} \quad (\text{S21})$$

$$\frac{d}{d\theta} \left(\frac{\kappa'_j}{\kappa'_i} \right) = s \left(\frac{\kappa_j}{\kappa_i} \right)^{s-1} \frac{d}{d\theta} \left(\frac{\kappa_j}{\kappa_i} \right). \quad (\text{S22})$$

Since ρ and κ_j/κ_i are always positive, directional sensitivities of ρ and κ_j/κ_i to parameter θ are preserved. Similarly, directional sensitivities of $1 - \rho$ to θ are preserved:

$$\frac{d(1 - \rho')}{d\theta} = s\rho^{s-1} \frac{d(1 - \rho)}{d\theta} \quad (\text{S23})$$

Therefore, we focus on comparisons of qualitative outcomes from SCT and refrain from making any comparisons of magnitude of niche and fitness difference values.

S3 Applications to classic two-strategy games

Consider a two-strategy game described by the following payoff matrix:

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (\text{S24})$$

There are four classic games that we can define from this simple payoff matrix: prisoner's dilemma ($T > R > P > S$), hawk-dove game ($T > R, S > P$), stag-hunt game ($R > T, P > S$), and harmony game ($R > T, S > P$). Here, we show that each game corresponds to a distinct region in the niche-fitness difference space under SCT.

To do so, we first begin by deriving expressions for niche overlap ρ and fitness difference κ_j/κ_i . Writing x_1 to denote the frequency of strategy 1, the expected payoff for each strategy corresponds to:

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (\text{S25})$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (\text{S26})$$

Then, multiplicative payoffs for each strategy when rare can be written as:

$$W_1(1) = \exp(R) \quad (\text{S27})$$

$$W_1(0) = \exp(S) \quad (\text{S28})$$

$$W_2(1) = \exp(T) \quad (\text{S29})$$

$$W_2(0) = \exp(P) \quad (\text{S30})$$

Therefore, we have:

$$\rho = \exp \left(\frac{R + P - S - T}{2} \right), \quad (\text{S31})$$

$$\frac{\kappa_2}{\kappa_1} = \exp \left(\frac{T + P - R - S}{2} \right). \quad (\text{S32})$$

482 As before, SCT allows us to predict the mutual invasion of two strategy using the
 483 following inequality:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S33})$$

484 Then, simple algebra reveals that each condition in mutual invasion inequality
 485 can be equivalently described by the payoff inequality:

$$\rho < \frac{\kappa_2}{\kappa_1} \iff R < T \quad (\text{S34})$$

$$\rho > \frac{\kappa_2}{\kappa_1} \iff R > T \quad (\text{S35})$$

$$\frac{\kappa_2}{\kappa_1} < \frac{1}{\rho} \iff P < S \quad (\text{S36})$$

$$\frac{\kappa_2}{\kappa_1} > \frac{1}{\rho} \iff P > S \quad (\text{S37})$$

486 These inequalities then allow us to classify outcomes of two-strategy games from the
 487 perspective of SCT. Note that the inequality between R and P does not affect the
 488 outcome.

489 S4 Applications to mechanisms for the evolution of coopera- 490 tion

491 We consider five mechanisms for the evolution of cooperation, all of which can be
 492 represented as a simple 2×2 payoff matrix (Nowak, 2006). Here, we go through each
 493 mechanism and derive measures for niche and fitness differences.

494 **Prisoner's dilemma.** Here, we consider a special case of Prisoner's dilemma
 495 which can be described by the following payoff matrix:

$$\begin{pmatrix} b - c & -c \\ b & 0 \end{pmatrix}, \quad (\text{S38})$$

496 where b represents the benefit for the recipient and c represents cost for the donor.
 497 In this case, the niche and fitness differences correspond to:

$$1 - \rho = 0 \quad (\text{S39})$$

$$\frac{\kappa_2}{\kappa_1} = \exp(c). \quad (\text{S40})$$

498 Therefore, as long as the cost is greater than zero, prisoner's dilemma always results
 499 in the competitive exclusion of cooperation.

500 **Kin selection.** Kin selection can be described by the following payoff matrix:

$$\begin{pmatrix} (b - c)(1 + r) & br - c \\ b - rc & 0 \end{pmatrix}, \quad (\text{S41})$$

where r represents the genetic relatedness. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 0 \quad (\text{S42})$$

$$\frac{\kappa_2}{\kappa_1} = \exp(c - br). \quad (\text{S43})$$

Therefore, defection will exclude cooperation when $c > br$. Equivalently, cooperation will be favored when $r > c/b$, which recovers the classic result.

Direct reciprocity. Direct reciprocity can be described by the following payoff matrix:

$$\begin{pmatrix} (b-c)/(1-w) & -c \\ b & 0 \end{pmatrix}, \quad (\text{S44})$$

where w represents the probability of next round. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 1 - \exp\left(\frac{(b-c)/(1-w) + c - b}{2}\right) \quad (\text{S45})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{b + c - (b-c)/(1-w)}{2}\right). \quad (\text{S46})$$

Therefore, defection will exclude cooperation when

$$\rho < \frac{\kappa_2}{\kappa_1} \quad (\text{S47})$$

$$\iff \exp\left(\frac{(b-c)/(1-w) + c - b}{2}\right) < \exp\left(\frac{b + c - (b-c)/(1-w)}{2}\right) \quad (\text{S48})$$

$$\iff bw < c \quad (\text{S49})$$

However, note that the following inequality holds whenever $c > 0$:

$$\frac{1}{\rho} < \frac{\kappa_2}{\kappa_1}. \quad (\text{S50})$$

Therefore, when $w > c/b$, we will have

$$\frac{1}{\rho} < \frac{\kappa_2}{\kappa_1} < \rho, \quad (\text{S51})$$

which implies priority effect (equivalently, bistability). Even as $w \rightarrow 1^-$, this inequality holds and therefore, cooperation cannot competitively exclude defection.

Indirect reciprocity. Indirect reciprocity can be described by the following payoff matrix:

$$\begin{pmatrix} b-c & -c(1-q) \\ b(1-q) & 0 \end{pmatrix}, \quad (\text{S52})$$

516 where q represents social acquaintanceship. In this case, the niche and fitness differ-
 517 ences correspond to:

$$1 - \rho = 1 - \exp\left(\frac{(b - c)q}{2}\right), \quad (\text{S53})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{2c - (b + c)q}{2}\right). \quad (\text{S54})$$

518 In this case, the condition for the dominance of defection (and therefore competitive
 519 exclusion of cooperation) can be written as:

$$\rho < \frac{\kappa_2}{\kappa_1} \quad (\text{S55})$$

$$\iff bq < c \quad (\text{S56})$$

520 Moreover, the following inequality always holds as long as $c > 0$.

$$\frac{1}{\rho} < \frac{\kappa_2}{\kappa_1}, \quad (\text{S57})$$

521 which implies that coexistence between cooperation and defection is not possible.
 522 Therefore, when $q < c/b$, then defection will competitively exclude cooperation.
 523 Otherwise, the system will enter the priority-effect regime.

524 Interestingly, when $q \rightarrow 1$, we have

$$\rho = \exp\left(\frac{b - c}{2}\right) \quad (\text{S58})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{c - b}{2}\right) \quad (\text{S59})$$

$$\frac{1}{\rho} = \exp\left(\frac{c - b}{2}\right) \quad (\text{S60})$$

$$(\text{S61})$$

525 In other words, the following inequality will hold:

$$\frac{1}{\rho} = \frac{\kappa_2}{\kappa_1} < \rho \quad (\text{S62})$$

526 **Network reciprocity.** Network reciprocity can be described by the following
 527 payoff matrix:

$$\begin{pmatrix} b - c & H - c \\ b - H & 0 \end{pmatrix}, \quad (\text{S63})$$

528 where $H = [(b - c)k - 2c]/[(k + 1)(k - 2)]$ and k represents the number of neighbors.
 529 In this case, the niche and fitness differences correspond to:

$$1 - \rho = 0, \quad (\text{S64})$$

$$\frac{\kappa_2}{\kappa_1} = \exp(-H + c), \quad (\text{S65})$$

530 which implies that there is complete niche overlap and that fitness difference decreases
 531 with H . Since H is a monotonically decreasing function of k when $k > 2$ and $b/c \geq 2$,
 532 fitness difference will increase with k . In particular, cooperation will competitively
 533 exclude defection when $H > c$, meaning that:

$$b/c > k. \quad (\text{S66})$$

534 **Group selection.** Group selection can be described by the following payoff
 535 matrix:

$$\begin{pmatrix} (b-c)(m+n) & (b-c)m - cn \\ bn & 0 \end{pmatrix}, \quad (\text{S67})$$

536 where m represents the number of groups and n represents the group size. In this
 537 case, the niche and fitness differences correspond to:

$$1 - \rho = 0, \quad (\text{S68})$$

$$\frac{\kappa_2}{\kappa_1} = \exp((c-b)m + cn), \quad (\text{S69})$$

538 which implies that there is complete niche overlap. The fitness difference will increase
 539 with n but decrease with m , provided that $b > c$. In particular, cooperation will
 540 competitively exclude defection under the following condition:

$$1 + \frac{n}{m} < \frac{b}{c}. \quad (\text{S70})$$

541 **S5 Applications to evolutionary games with envi-** 542 **ronmental stochasticity**

543 Consider a competition of two strategies under stochastic payoffs:

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix} + \frac{\zeta}{\sqrt{s}} \begin{pmatrix} \sigma_R & \sigma_S \\ \sigma_T & \sigma_P \end{pmatrix}, \quad (\text{S71})$$

544 Assuming weak selection $s \ll 1$, the evolution of two competing strategies in a large
 545 population can be described by the following equation:

$$\frac{dx}{dt} = sx(1-x) \left(\pi_C - \pi_D + \left(\frac{1}{2} - x \right) (\sigma_C - \sigma_D)^2 \right), \quad (\text{S72})$$

546 To simplify, we write

$$\frac{dx}{dt} = sx(1-x)G(x), \quad (\text{S73})$$

547 where

$$G(x) = \left(\pi_C - \pi_D + \left(\frac{1}{2} - x \right) (\sigma_C - \sigma_D)^2 \right). \quad (\text{S74})$$

Thus, the sign of $G(x)$ at boundary conditions determines whether a strategy can invade when the competing strategy is at equilibrium.

Note that we cannot immediately apply the original SCT presented in the main text here because $G(x)$ does not uniquely separate into two payoff measures. Instead, we can still utilize the invasion growth rate when rare and follow the same procedure. In particular, the mutual invasion condition at monomorphic boundary can be written as:

$$0 < G(0) \quad (\text{S75})$$

$$0 < -G(1) \quad (\text{S76})$$

Exponentiating both, we can combine them into a single inequality:

$$\exp(G(1)) < 1 < \exp(G(0)) \quad (\text{S77})$$

Dividing both sides by $\exp((G(1) + G(0))/2)$, we get:

$$\exp\left(\frac{G(1) - G(0)}{2}\right) < \exp\left(-\frac{G(1) + G(0)}{2}\right) < \exp\left(\frac{G(0) - G(1)}{2}\right), \quad (\text{S78})$$

which gives us expressions for niche overlap and fitness difference:

$$\rho = \exp\left(\frac{G(1) - G(0)}{2}\right) \quad (\text{S79})$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(-\frac{G(1) + G(0)}{2}\right) \quad (\text{S80})$$

Then, by substituting expressions for $G(0)$ and $G(1)$, we obtain following expressions for niche and fitness difference:

$$1 - \rho = 1 - \exp\left(\frac{R + P - S - T}{2} - \frac{(\sigma_R - \sigma_T)^2 + (\sigma_S - \sigma_P)^2}{4}\right) \quad (\text{S81})$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(\frac{T + P - R - S}{2} + \frac{(\sigma_R - \sigma_T)^2 - (\sigma_S - \sigma_P)^2}{4}\right) \quad (\text{S82})$$

Now, to study the interior behavior of the system, we assume that the amount of stochasticity is proportional to deterministic payoffs (Wang et al., 2025):

$$\begin{pmatrix} \sigma_R & \sigma_S \\ \sigma_T & \sigma_P \end{pmatrix} = k \begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (\text{S83})$$

562 Then, the expression for $G(x)$ can be written as:

$$G(x) = ((R - T)x + (S - P)(1 - x)) \times \quad (\text{S84})$$

$$\left(1 + k^2 \left(\frac{1}{2} - x\right) ((R - T)x + (S - P)(1 - x))\right) \quad (\text{S85})$$

$$= ((R + P - T - S)x + (S - P)) \times \quad (\text{S86})$$

$$\left(1 + k^2 \left(\frac{1}{2} - x\right) ((R + P - T - S)x + (S - P))\right) \quad (\text{S87})$$

$$= k^2(R + P - T - S)^2 (x - x^*) \left(K + \left(\frac{1}{2} - x\right) (x - x^*)\right), \quad (\text{S88})$$

563 where $x^* = (P - S)/(R + P - T - S)$ and $K = 1/(k^2(R + P - T - S))$. Therefore,
564 this system has internal equilibrium $x = x^*$ when $0 < (P - S)/(R + P - T - S) < 1$. It
565 can also have two additional equilibrium when $K + \left(\frac{1}{2} - x\right) (x - x^*) = 0$ has positive,
566 real roots. In particular, the latter equation can be written as:

$$x^2 - \left(\frac{1}{2} + x^*\right)x + \frac{x^*}{2} - K = 0. \quad (\text{S89})$$

567 Thus, two additional equilibrium correspond to:

$$x = \frac{x^* + \frac{1}{2} \pm \sqrt{\left(\frac{1}{2} - x^*\right)^2 + 4K}}{2}. \quad (\text{S90})$$

568 This shows that two additional equilibrium exists when

$$\left(\frac{1}{2} - x^*\right)^2 + 4K > 0. \quad (\text{S91})$$

569 Note that these equilibrium may not necessarily fall between 0 and 1.

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